PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

HEADLESS CLOUD-NATIVE SEQUENCER WITH LATENCY-COMPENSATED DISTRIBUTED DAW SYNCHRONIZATION

Title of the Invention	Headless Cloud-Native Sequencer with Latency-Compensated Distributed DAW Synchronization
Docket Number	HS-2026-003
Inventor(s)	Eric Haywood
Residence(s)	149 Remington Street Unit 425, Fort Collins, CO 80524
Correspondence Address	HeadySystems Inc., 149 Remington Street Unit 425, Fort Collins, CO 80524
Attorney/Agent (Name & Reg. No.)	None
U.S. Government Interest (Yes/No)	No
Gov. Agency & Contract No.	N/A

Title of the Invention

[0001] Headless Cloud-Native Sequencer with Latency-Compensated Distributed DAW Synchronization

Cross-Reference to Related Applications

[0002] This application is related to co-pending provisional applications Docket Nos. HS-2026-001 and HS-2026-002, filed concurrently herewith by the same inventor and assignee.

Field of the Invention

[0003] The invention relates to real-time multimedia synchronisation over computer networks, and more particularly to a headless cloud-native MIDI sequencer acting as an absolute master clock for globally distributed live performance with sub-millisecond timeline fidelity.

Background of the Invention

[0004] Live musical performance across geographically distributed participants requires precise temporal synchronisation. TCP's guaranteed delivery mechanism introduces queueing delays incompatible with real-time music, where latency must be below the perceptual threshold. Sound travels at about 1 foot per millisecond; two musicians 15 feet apart experience roughly 15 ms latency, which is tolerable[1]. User studies indicate that one-way audio delays above about 40 ms disrupt performance, whereas delays below 40 ms are generally acceptable[2].

Summary of the Invention

[0005] The invention provides a cloud-native sequencer that serves as the absolute master clock for distributed digital audio workstations (DAWs). The sequencer maintains microsecond-precision timing, broadcasts the master clock to participants via low-latency UDP, and compensates for network jitter and latency. Clients calculate round-trip latency and adjust playback positions accordingly. A jitter buffer ensures that audio and MIDI messages remain aligned within the 40 ms perceptual window.

Detailed Description of the Preferred Embodiment

A. Master Clock Architecture

[0006] The sequencer runs in a headless cloud environment with high-precision timers. It publishes the current tick via a gRPC and WebSocket interface. Clients subscribe and receive periodic clock packets containing the current tick and scheduled future ticks.

B. Latency Measurement and Compensation

[0007] Clients measure round-trip latency to the master clock using ping messages. Each client adjusts its local playback position by subtracting half of the round-trip time. The system monitors latency drift and applies small adjustments to keep all clients synchronised within ±5 ms.

C. Jitter Buffering and Tolerable Latency

[0008] A jitter buffer holds incoming MIDI events for a configurable duration (e.g., 30 ms) to absorb network variation. Because delays above 40 ms become disruptive for musicians[2], the system ensures that the sum of network latency and jitter buffer stays below this threshold. Additional strategies include forward error correction and redundancy to mitigate packet loss.

References

1. [1] The latency of sound in air is roughly 1 ms per foot; two musicians 15 feet apart experience about 15 ms latency, which is tolerable for live performance【263500279240673†L42-L49】.
2. [2] User studies show that network music becomes disruptive when one-way audio delay exceeds about 40 ms; delays below 40 ms are generally acceptable【253068863342350†L1096-L1123】.

Application Data Sheet 37 CFR 1.76

The Application Data Sheet is part of the provisional or non-provisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76.

Inventor Information

of inventors: 1

1. **Eric Haywood**

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Application Information

Customer number	**221639 -**
Correspondence address	---
Title of invention	**HEADLESS CLOUD-NATIVE SEQUENCER WITH LATENCY-COMPENSATED DISTRIBUTED DAW SYNCHRONIZATION**
Attorney docket number	**---**
Entity status	---

Application type	Provisional Application under 35 USC 111(b)
Subject matter	Utility
Total number of drawing sheets	---
Filing by reference	No

Representative Information

of representatives: 0

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32).

Data was not provided for this section.

Domestic Benefit/National Stage Information

of benefit claims: 0

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c), 386(c), or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Data was not provided for this section.

Foreign Priority Information

of foreign priority claims: 0

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the Application Data Sheet constitutes the claim for priority as required by 35 U.S.C. 119 (b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a

copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Data was not provided for this section.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Checking this box will cause the application to be examined under the first inventor to file provisions of the AIA.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 2016, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization or Opt-Out of Authorization to Permit Access

When this Application Data Sheet is properly signed and filled with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE:
This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. **Priority Document Exchange (PDX)**
Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the China National Intellectual Property Administration (CNIPA), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its

related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h) (1).

B. **Search Results from U.S. Application to EPO**

Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with the search results from the instant application.

NOTE:

Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Applicant Information

of applicants: 0

The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46.

Data was not provided for this section.

Assignee Information including Non-Applicant Assignee Information

of assignees: 1

An assignee-applicant identified in the "Applicant" section will appear on the patent application as an applicant.

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

HeadySystems Inc.
149 Remington Street Unit 425
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UNITED STATES

9704445112

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admin@headysystems.com

Organization

Signature

NOTE:

This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c)**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See CFR 1.4(d) for the manner of making signatures and certifications.

Signature	First name	Last name	Registration #	Date
/Eric Haywood/	ERIC	HAYWOOD	---	03/03/2026

UNITED STATES
PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
63/995,272	**03/03/2026 01:31:01 PM Z ET**	**-**

Title of Invention

HEADLESS CLOUD-NATIVE SEQUENCER WITH LATENCY-COMPENSATED DISTRIBUTED DAW SYNCHRONIZATION

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
CONFIRMATION #	1678	FILED BY	ERIC HAYWOOD
PATENT CENTER #	74696307	FILING DATE	-
CUSTOMER #	221639	FIRST NAMED INVENTOR	Eric Haywood
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
generatedADS74696307.pdf	5	Application Data Sheet	109 KB
HS-2026-003-Final-Assigned-Updated-SPEC.docx	3	Specification	38 KB

Warning: One or more pages are missing page numbering. Page numbering will be automatically applied after submission.

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
generatedADS74696307.pdf	FAEF61E89F95B39953E27434E3FA5DBD780429D1353A1CF9D

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54F63720F06DC591F0AE0A49EAF4CEF

HS-2026-003-Final-Assigned-Updated-SPEC.docx

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053FEBAEB687689DCF9DF13C5ED90C7E99F7FB7B1D6EA19
D528AB3E5C827798EB6B6D31C60A2F86

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.